

WHITE-TAILED DEER IMPACT ON THE VEGETATION DYNAMICS OF A NORTHERN HARDWOOD FOREST

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Abstract. Considerable controversy has arisen over the management of white-tailed deer in eastern landscapes where there is evidence of damage to forest vegetation, crops, and wildlife habitat attributable to deer. We examined the impact of 4, 8, 15, and 25 deer/km² on herbaceous layer abundance and tree seedling density, height development, species composition, and diversity for 10 yr in a repeated-measures randomized-complete-block experiment at four replicate cherry–maple forest sites in northwestern Pennsylvania, USA. At each 65-ha site, deer were placed in 13- or 26-ha fenced enclosures in which the landscape composition and forage production of a forest managed on a 100-yr rotation was simulated by clear-cutting 10% of each area and thinning 30%. Vegetation was sampled 0–1, 3, 5, and 10 yr after treatment (YAT). We analyzed vegetative treatment responses separately in each silvicultural treatment.

Time was an important factor mediating responses at all deer densities. *Rubus* spp. abundance in all silvicultural treatments and the density of striped maple (STM) in clearcuts and birch (BI), American beech (AB), and red maple (RM) in thinnings showed negative trends with increasing deer density. As deer density increased, we found negative linear trends in the height of BI, RM, and AB in clearcuts; in the height of AB, RM, and STM in thinnings; and in the height of AB and RM in uncut stands. Species richness showed a negative linear trend with increasing deer density in all silvicultural treatments.

Conversely, species avoided by deer or resilient to deer browsing increased with increasing deer density. Percent cover of ferns, grasses, and sedges showed positive linear trends with deer density in clearcuts and in thinnings. The percentage of plots dominated by black cherry (BC) increased with increasing deer density in cut and uncut stands. The

